CS543 · WEB SERVICES · ASSIGNMENT 8 OF 12

Build one CampusEats service as a REST API

Team submission

Goal. Take one small service from CampusEats and expose it as a **REST API** with Node & Express — list, fetch-one, and create — then demonstrate it with a simple HTML/CSS page.

TASK

Pick any one CampusEats entity — e.g. **Menu / Catalogue** (`id, name, price, available`), **Restaurants**, or **Orders**. Build its REST routes and a front-end. Follow the Catalogue tutorial as your template.

#	Do this
1	**Routes** — Define `GET /items` (list), `GET /items/:id` (one, with a **404** if absent), and `POST /items` (create, with a **201**). Use the right HTTP method and status code for each outcome.
2	**Data** — An in-memory array with 3–4 seed records as the source of truth. (A real database is Lecture 10; it is optional here.)
3	**Handlers** — Write each route. Read the id from `req.params.id` and the posted JSON from `req.body` (add `express.json()`). Validate the POST — reply **422** if a required field is missing.
4	**Front-end** — A plain HTML/CSS page that `fetch()`es the list and shows it, plus one form that POSTs a new record and refreshes the list. Serve it with `express.static('public')`.
5	**Run it** — Start the server, open the page, add one record live, and screenshot the before / after.

THE MINIMUM ROUTES

```
list, then add one
app.get('/items', (req, res) => res.json(items));

app.post('/items', (req, res) => {
  const { name } = req.body;
  if (!name) return res.status(422).json({ error: 'name required' });
  items.push({ id: nextId++, ...req.body });
  res.status(201).json(items.at(-1));
});
```

DELIVERABLES

- `server.js` — your Express REST API with working routes.
- `package.json` — with `express` listed as a dependency.
- `public/index.html` + `style.css` — the page that reads and writes the data.
- A **GitHub repo** with the above + a **Google Drive link** with a demonstration video + two screenshots (before / after the POST) and a one-line README on how to run it.
- One submission per team via Google Classroom — repo link, with roll numbers and names on the README.

ANSWER IN THE README

1. Which CampusEats service did you pick, and what are its routes (method + path)?

2. Which **status codes** did you return, and for which outcomes (200 / 201 / 404 / 422)?

3. Where did you use `req.params` and where `req.body`? Why does POST need `express.json()`?

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